

Plan of work — data & upload architecture

O&M Cane Training · teacher-facing O&M assessment app for blind / visually-impaired children

Prepared for external developer review. Captures the decision in play, the two candidate architectures, what is settled vs. open, and the planned build path for each.

CONTEXT IN ONE PARAGRAPH

The app runs structured O&M assessment sessions and records results. It is plain HTML/CSS/JS, no bundler (a deliberate choice to keep a no-coder content-edit workflow), wrapped natively with Capacitor 8, Android-first. It already builds and runs on an emulator; local storage via Capacitor Preferences is durable across restarts; CSV export works; child profiles and session records persist; the repo is on GitHub. Storage goes through a single **Store** seam with a write-verify guard. The next major capability is sending **session video** to the cloud (India-region Supabase) so the R&D team (NCAHT / IIT Delhi) can use it. This report is about the shape that data movement should take.

THE DECISION IN PLAY

How should session data (including face-visible video of children) leave the device and reach the R&D team? Two candidate shapes. The choice hinges on one question we are confirming with the R&D team: **do they want to analyse data across many children as a dataset, or watch individual sessions to improve the assessment procedure?**

	Option A — Central corpus	Option B — Reviewable session archive
What it is	All records merged into one child-keyed dataset, queryable across teachers.	A bucket of individual sessions; each = video + activity + result + notes.
Researcher does	Runs queries / statistics across the child population.	Opens and watches individual sessions to diagnose the procedure.
Fits which purpose	Longitudinal / cohort analysis, outcomes by activity at population scale.	Reviewing sessions to find where assessment breaks & what activities to add.
Must be built	Cross-teacher child-identity resolution, dedup/merge, one-record-per-child join, query layer.	Upload session + metadata; review access for the institution. No merge/dedup/join.
Relative build cost	Higher.	Lower.
Identified-video duties	Access control, encryption, delete-on-withdrawal.	Identical — same duties; identified video sets the bar regardless of shape.

Current lean: if the R&D team's answer is 'watch sessions to improve the tool' (which is what they have described so far), **Option B** is the better fit — it does what the purpose needs without the corpus machinery. Final confirmation from R&D is pending.

WHAT IS ALREADY SETTLED (regardless of A or B)

1. **Stamp childId + teacherId on every record from v1.** Cheap now; it is the spine of Option A and the upgrade path from B→A later. Skipping it turns a future sync into a data-archaeology migration.
2. **Identified video is confirmed needed.** R&D need to see the child (orientation, head-turning, cane technique), so faces are in scope. This is the highest-sensitivity version and sets non-negotiable handling duties.
3. **Three duties live in the upload path from day one, not deferred:** access control (who can open video), encryption, and delete-on-withdrawal. These are not solvable by consent text and are cheapest built in from the start.
4. **Sequencing:** local-first ships first; the cloud data side activates once the data-sharing agreement and consent are in place. Choosing the shape now just means building the local data model already shaped for it.

WHAT IS STILL OPEN

- a. **The A-vs-B question** — pending R&D confirmation (query a dataset vs. watch sessions).
- b. **Consent text & data-sharing agreement** — must name IIT Delhi / NCAHT and the purpose (improving the assessment via identified video). Required before any real child data flows; can be drafted in parallel with the build.
- c. **Offline-classroom upload behaviour** — immediate vs. queue-and-upload-later. Affects the uploader design; not yet decided.

PLANNED BUILD PATH (once shape is confirmed)

Both shapes share the same first moves; they diverge only at the merge/query layer, which only Option A needs.

- 1 Define the **session record**: fields that ride with each video (childId, teacherId, activity, result, teacher notes, timestamp) so a reviewer has full context.
- 2 Add an **Uploader seam** beside the existing Store seam — same abstraction discipline, so the cloud target can be swapped/tested in isolation.
- 3 Wire video + metadata upload to India-region Supabase bucket, with access control + encryption + a delete-on-withdrawal path built in.
- 4 Resolve offline behaviour (immediate vs. queue) and implement the chosen one in the Uploader.
- 5 **Option A only:** add child-identity resolution, dedup/merge, and the cross-teacher query layer.
- 6 Headless-verify the upload chain (write / read-back / verify) before shipping, consistent with current test discipline.

QUESTIONS FOR THE REVIEWER

- Does the A-vs-B framing hold, or is there a third shape we are missing for 'researchers review identified video to improve a tool'?
- Supabase bucket + Postgres for this, or would you architect the identified-video store differently (signed URLs, row-level security, separate media store)?

- Offline upload: immediate or queue-and-retry — any strong opinion given intermittent classroom connectivity?
- Anything in the 'settled' list you would challenge — particularly the childId/teacherId stamping and the day-one handling duties?

This is an engineering plan-of-work, not legal advice. The consent text and the app-to-institution data-sharing agreement should be reviewed by someone qualified before any real child data is collected. The app handles children's data under India's DPDP Act 2023 + Rules 2025.